

ElectViz Data Visualization for Media

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Project Overview



Purpose

Transform complex election data into clear, insightful visual stories that resonate with diverse audiences



Scope

National and regional election results, voter demographics, and comprehensive trend analysis



Goal

Enhance understanding and engagement through interactive and static visualizations

Data Sources

The dataset provides detailed information on election candidates, results, and performance across all Indian states. It enables both **macro-level** (state/national) and **micro-level** (constituency/candidate) analysis for visualization.

st_name	year	ac_no	ac_name	ac_type	cand_name	cand_sex	partyname	partyabbre	totvotpoll	electors
Maharashtra	1985	1		GEN	Balkrishna Raghoba Sawant	M	Independent	IND	1534	111887
Maharashtra	1985	1		GEN	Vijay Kerkar	M	Independent	IND	1776	111887
Maharashtra	1985	1		GEN	Vasant Kesarkar	M	Independent	IND	6276	111887
Maharashtra	1985	2		GEN	Vilas Shantaram Tari	M	Independent	IND	246	104559
Maharashtra	1985	2		GEN	Dattatray Narayan Kamat	M	Independent	IND	307	104559
Maharashtra	1985	2		GEN	Narayan Vasudeo Pitre	M	Independent	IND	670	104559
Maharashtra	1985	2		GEN	C. S. Gaonkar	M	Independent	IND	858	104559
Maharashtra	1985	2		GEN	Vasantrao Khot	M	Independent	IND	1786	104559
Maharashtra	1985	2		GEN	Jagannath Ramachandra Ghatwal	M	Independent	IND	2682	104559
Maharashtra	1985	2		GEN	S. N. Desai	M	Independent	IND	8736	104559
Maharashtra	1985	3		GEN	Amrut Raghunath Parab	M	Independent	IND	175	100743
Maharashtra	1985	3		GEN	Vasantrao Dhanaji Moperkar	M	Independent	IND	189	100743
Maharashtra	1985	3		GEN	Bhai Chindarkar	M	Independent	IND	791	100743
Maharashtra	1985	3		GEN	Appasaheb Chavan	M	Independent	IND	2331	100743
Maharashtra	1985	3		GEN	Kalashkar Vinaykand Appant	M	Independent	IND	2171	100743

PC Name	Winning Candidate	Winning Party	Runner-up Canddiate	Runner-up Party	Margin Vote
Arunachal East	TAPIR GAO	Bharatiya Janata Party	BOSIRAM SIRAM	Indian National Congress	30421
Arunachal West	KIREN RIJU	Bharatiya Janata Party	NABAM TUKI	Indian National Congress	100738
Darrang-Udalguri	DILIP SAIKIA	Bharatiya Janata Party	MADHAB RAJBANGSHI	Indian National Congress	329012
Guwahati	BIJULI KALITA MEDHI	Bharatiya Janata Party	MIRA BORTHAKUR GOSWAMI	Indian National Congress	251090
Karimganj	KRIPANATH MALLAH	Bharatiya Janata Party	HAFIZ RASHID AHMED CHOUDHURY	Indian National Congress	18360
Kaziranga	KAMAKHYA PRASAD TASA	Bharatiya Janata Party	ROSELINA TIRKEY	Indian National Congress	248947
Lakhimpur	PRADAN BARUAH	Bharatiya Janata Party	UDAY SHANKAR HAZARIKA	Indian National Congress	201257
Silchar	PARIMAL SUKLABAIDYA	Bharatiya Janata Party	SURYA KANTA SARKAR	Indian National Congress	264311
Sonitpur	RANJIT DUTTA	Bharatiya Janata Party	PREMLAL GANJU	Indian National Congress	361408
Maharajganj	JANARDAN SINGH "SIGRIWAL"	Bharatiya Janata Party	AAKASH KUMAR SINGH	Indian National Congress	102651
Muzaffarpur	RAJ BHUSHAN CHOUDHARY	Bharatiya Janata Party	AJAY NISHAD	Indian National Congress	234927
Paschim Champaran	DR.SANJAY JAISWAL	Bharatiya Janata Party	MADAN MOHAN TIWARI	Indian National Congress	136568
Patna Sahib	RAVI SHANKAR PRASAD	Bharatiya Janata Party	ANSHUL AVIJIT	Indian National Congress	153846

Visualization Tools & Techniques



Interactive Electoral Maps

State and county-level results with dynamic exploration capabilities



Statistical Charts

Bar and pie charts visualizing vote shares and demographic breakdowns



Trend Analysis

Time series graphs displaying turnout patterns across multiple elections



Color Coding System

Intuitive red/blue scheme for immediate political party recognition

Data Visualization Examples

Card-Based Visualizations

Interactive cards displaying individual candidate or constituency data with key metrics

Waterfall Model Charts

Step-by-step visualization showing how vote counts flow and change through different stages

Trend Analysis Graphs

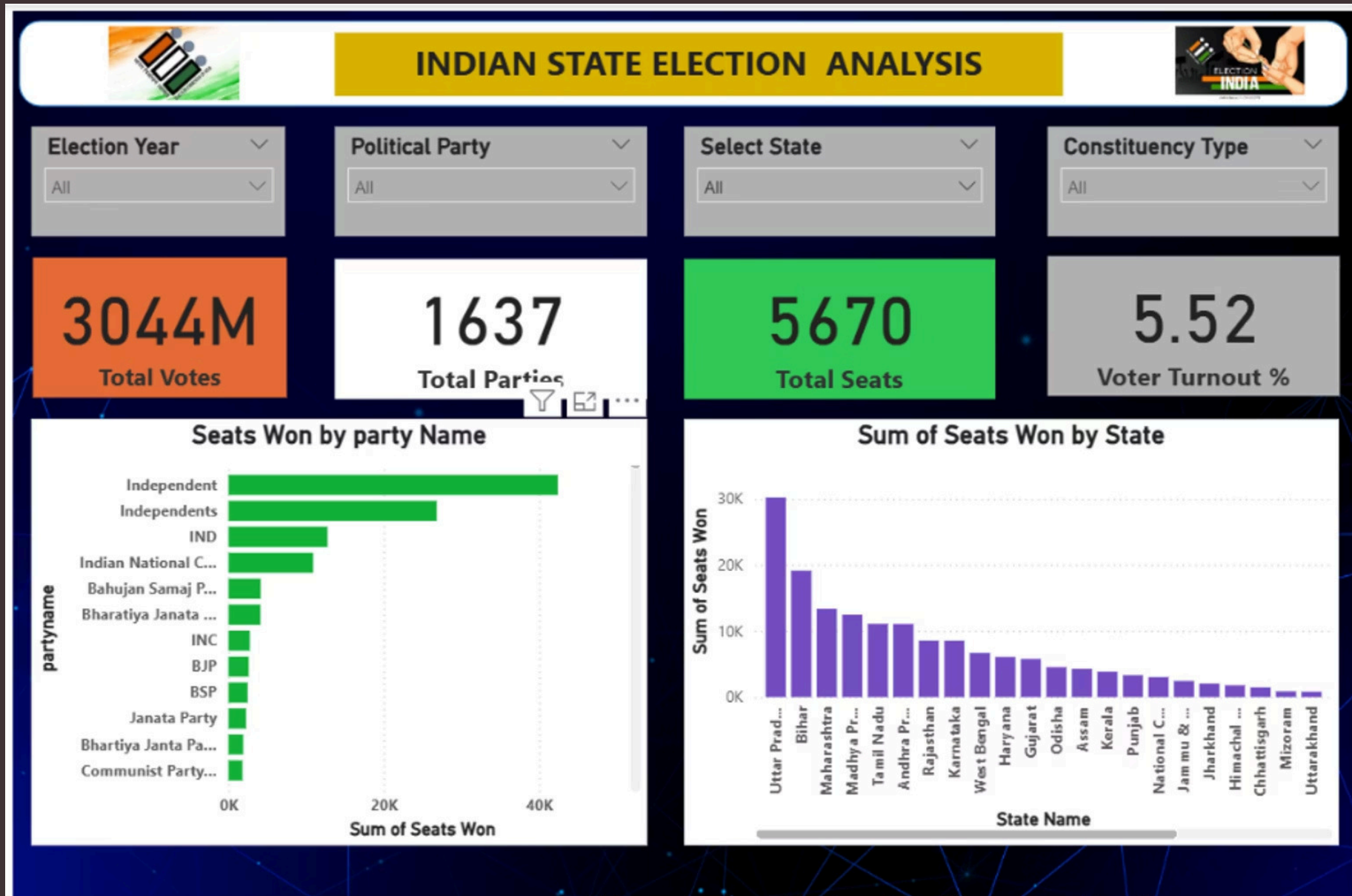
Track and analyze election patterns and voter behavior over time using line graphs

Detailed Statistical Views

Access in-depth statistics and vote counts for specific data points within any visualization



Snapshots of Visualizations



Top Section:

- Displays key figures — **Total Votes (3044M)**, **Total Parties (1637)**, **Total Seats (5670)**, and **Voter Turnout (5.52%)**
- Gives a quick snapshot of the election scale and participation

Bar Chart (Left – Seats Won by Party Name):

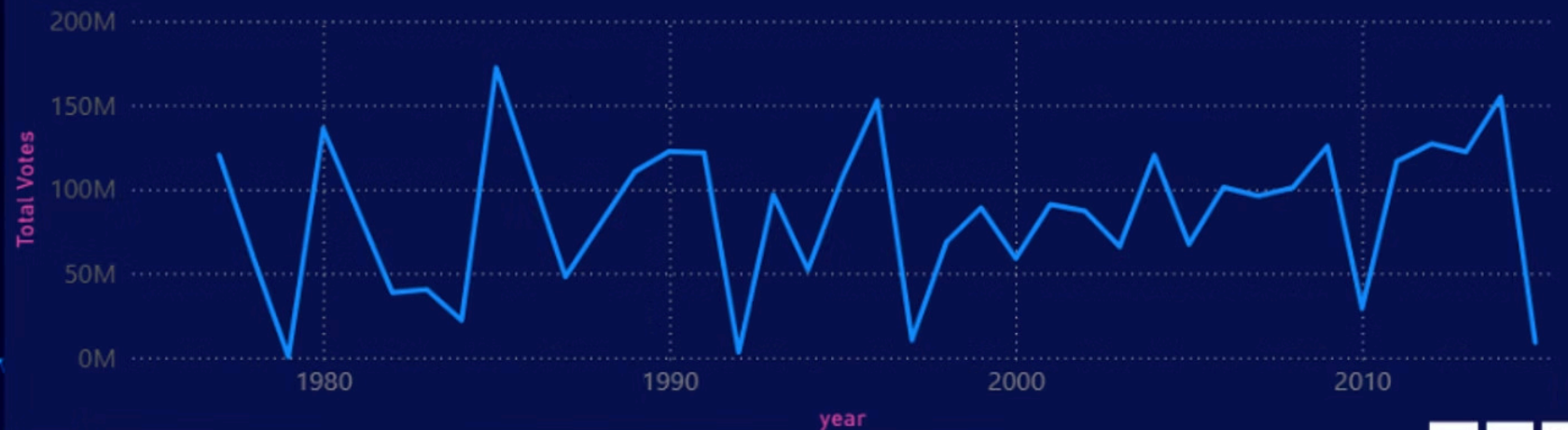
- Shows how many seats each political party has won
- Independent candidates dominate, followed by parties like INC, BJP, BSP, and others

Bar Chart (Right – Seats Won by State):

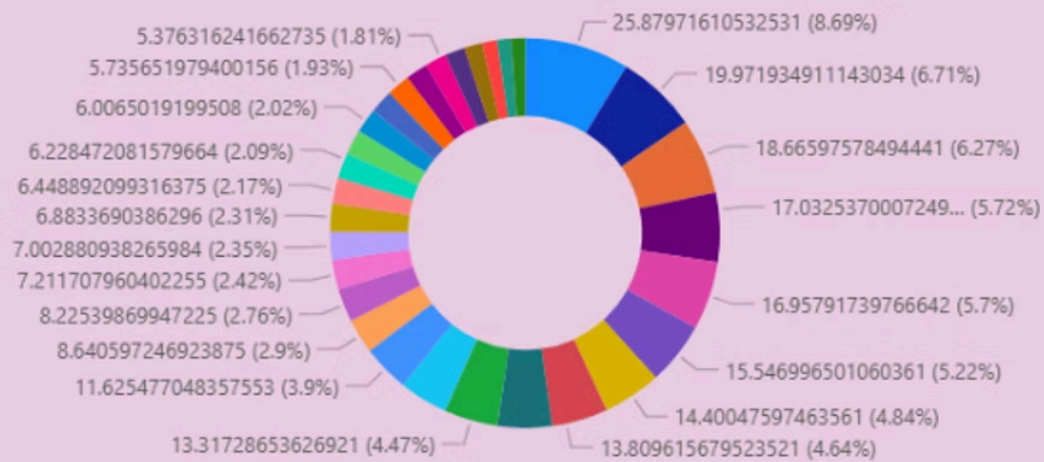
- Displays the total seats won across different states
- States like **Uttar Pradesh**, **Bihar**, and **Maharashtra** have the highest number of seats
- Reflects their large assemblies and voter bases

Overall, this dashboard helps understand **party performance** and **state-wise seat distribution** at a glance.

Total Votes by year



Voter Turnout % by State Name



State Name

- Nagaland
- Tripura
- Arunachal Pradesh
- Manipur
- Mizoram
- Meghalaya
- Goa
- Puducherry
- Sikkim

Left Chart (Count of Seats Won by State Name)

This horizontal bar chart ranks states by the total number of seats won across all parties. Uttar Pradesh leads with the highest share (100%), followed by Bihar, Maharashtra, Madhya Pradesh, Andhra Pradesh, Tamil Nadu, Rajasthan, Karnataka, West Bengal, and Haryana. Smaller states like Sikkim, Tripura, Goa, Nagaland, and Puducherry contribute less than 1%. Interpretation: Larger states dominate due to more constituencies, while smaller regions naturally have fewer seats.

Right Chart (Vote Share by Party)

This pie chart shows the proportion of seats won by each political party across India. Independent candidates hold the largest share (84K seats), followed by INC (53K), BJP (25K), and BSP (22K). Other national and regional parties contribute smaller shares. Interpretation: Independent candidates show strong nationwide presence, while major parties hold significant but smaller portions of total seats.

Line Chart (Total Votes by Year)

- Shows how the total number of votes has changed over time.
- The ups and downs reflect changes in voter participation across different election years.

Donut Chart (Voter Turnout % by State)

- Represents the percentage of voter turnout across various states.
- States like Nagaland (8.69%), Tripura (6.71%), and Arunachal Pradesh (6.27%) show higher turnout rates compared to others.
- These visuals highlight how voting behavior has evolved over time and which states show stronger voter participation.

Top Party

Independent

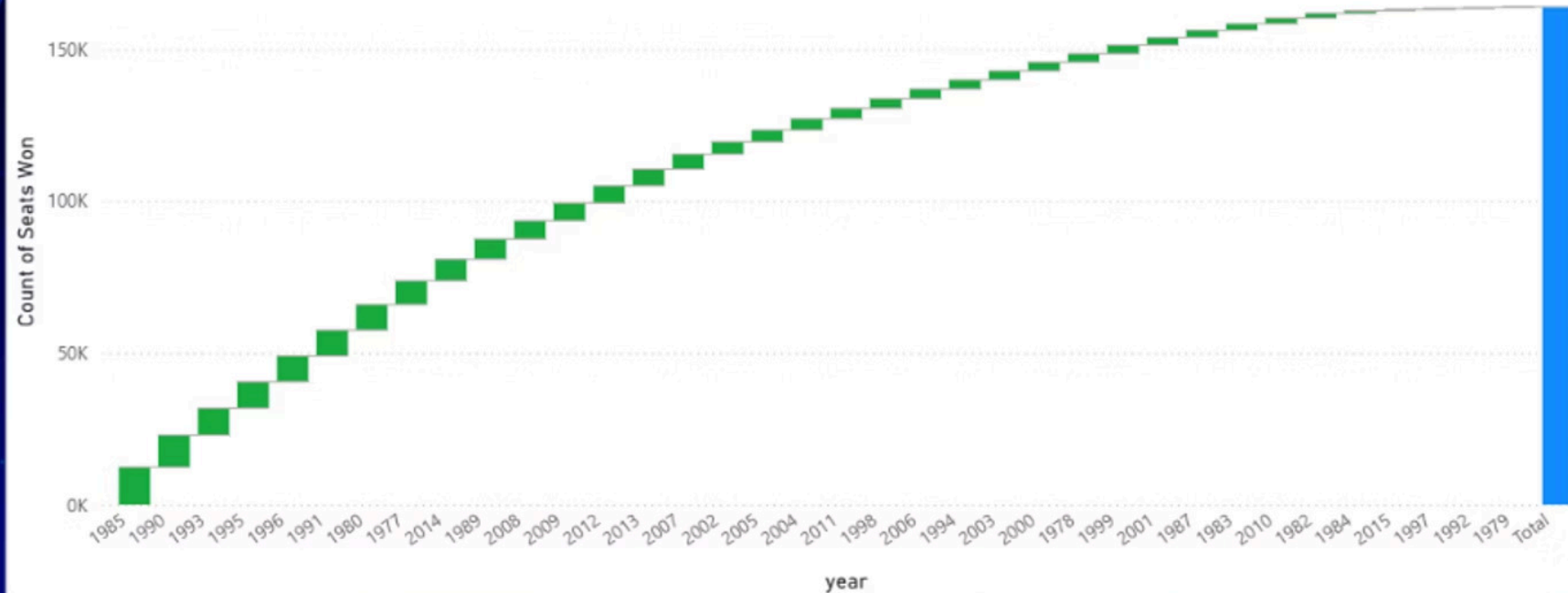
Top State

Uttar Pradesh



Count of Seats Won by year

● Increase ● Decrease ● Total



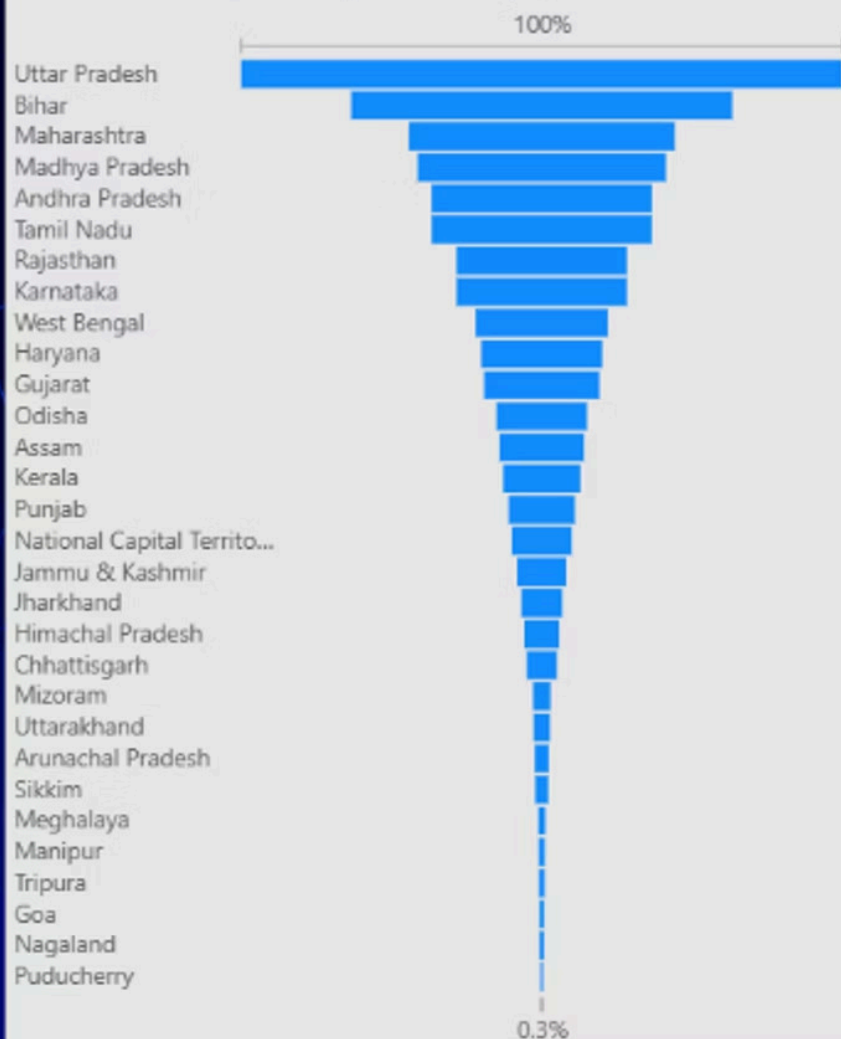
Bar Chart (Count of Seats Won by Year):

- Displays how the total number of seats won has changed across different election years (from 1960 to 2020+)
- Green bars indicate an increase in seats compared to the previous year
- Blue bars represent the overall seat count across the timeline

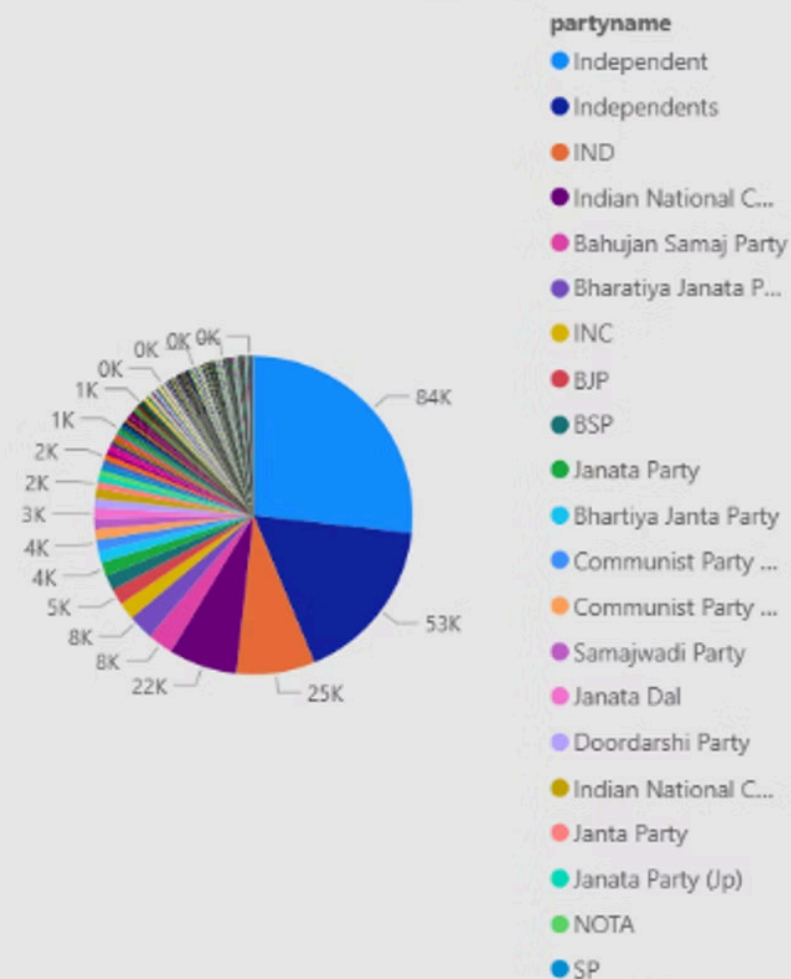
Cards:

- Top Party – Independent: Independent candidates have collectively won the most seats across the dataset
- Top State – Uttar Pradesh: Leads with the highest number of seats won due to its vast number of constituencies

Count of Seats Won by State name



Vote share by party



Left Chart (Count of Seats Won by State Name):

- Horizontal bar chart ranks states by total number of seats won across all parties
- Uttar Pradesh leads with the highest share (100%), followed by Bihar, Maharashtra, Madhya Pradesh
- Smaller states like Sikkim, Tripura, Goa, Nagaland, and Puducherry contribute less than 1%
- Larger states dominate due to more constituencies, while smaller regions naturally have fewer seats

Right Chart (Vote Share by Party):

- Pie chart shows proportion of seats won by each political party across India
- Independent candidates hold the largest share (~84K seats), followed by INC (~53K), BJP (~25K), and BSP (~22K)
- Other national and regional parties contribute smaller shares
- Independent candidates show strong nationwide presence, while major parties hold significant but smaller portions of total seats

Sate-wise Party Seat Distribution

State Name	A S P	Aa S P	AAAP	AACP	Aadivasi Sena Party	AAMP	AAP	AASAP	AaSP	ABAS	ABCD(A)	ABDBM	ABGL	ABGP	ABHM	AB
Andhra Pradesh			79													
Arunachal Pradesh																
Assam																
Bihar																
Chhattisgarh		1												2	1	
Goa																
Gujarat					2											
Haryana																
Himachal Pradesh																
Jammu & Kashmir																
Jharkhand						1									2	
Karnataka															1	
Kerala																
Madhya Pradesh									8					16	3	
Maharashtra															11	
Manipur																
Meghalaya																
Mizoram																
Nagaland																
National Capital Territory Of Delhi		2	59	4				1							5	
Odisha			57													
Puducherry																
Punjab																
Rajasthan							2				4				11	
Sikkim																
Tamil Nadu															6	
Tripura																
Uttar Pradesh	4	1								1		1			6	
Total	4	4	195	4	2	1	2	1	8	1	4	1	2	18	47	

Major Seat Holders:

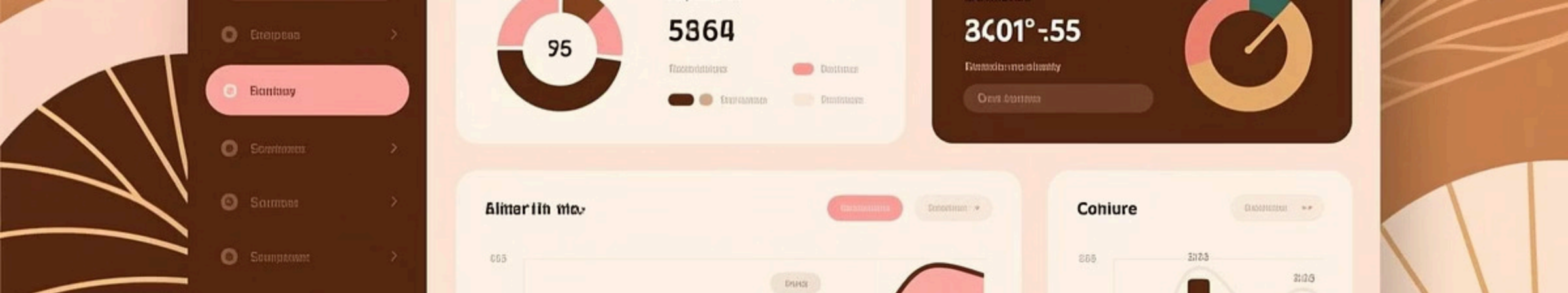
- AAP leads with 195 seats, showing strong influence in Delhi (59), Punjab (57), and Andhra Pradesh (79)
- Marks AAP's dominance in these key regions

State Highlights:

- Andhra Pradesh: AAP – 79 seats
- Delhi: AAP – 59; AASP (4), ABHM (5)
- Odisha: AAP – 57
- Madhya Pradesh: AACP (8), ABGP (16), ABHM (3)
- Rajasthan: ABCD(A) (2), ABGL (4), ABGP (11)
- Uttar Pradesh: AASP (4), AAP (1), ABGL (2), ABGP (18)

Minor Parties' Activity:

- Parties like ABGL, ABGP, ABHM hold scattered seats
- Reflects regional or issue-based influence in select states



Key Insights from Visualizations



Voter Turnout Trends

Analysis reveals increasing participation among younger demographics while certain regions show declining engagement patterns



Party Performance Shifts

Comparative data highlights significant changes in party strongholds and emerging competitive districts



Demographic Influences

Clear correlations between age, education, location, and voting preferences inform campaign strategies



Political Hotspots

Identification of rapidly shifting electoral landscapes and emerging battleground regions

Challenges Encountered

1

Data Quality Issues

Inconsistencies and missing values across multiple sources required extensive cleaning and validation

2

Design Balance

Maintaining optimal detail while ensuring clarity and preventing information overload

3

Technical Complexity

Building responsive interactive elements that perform smoothly across devices

4

Accessibility

Ensuring usability for diverse audiences including those with visual or cognitive differences



Impact & Applications



Journalism & Analysis

Supports media professionals in delivering accurate, compelling election coverage with visual evidence and clear narratives



Voter Empowerment

Provides citizens with transparent, accessible data to make informed decisions and understand their electoral landscape



Education

Empowers educators to teach electoral processes through engaging, interactive visualizations that make civics come alive



Forecasting Models

Establishes foundation for predictive analytics and future election modeling capabilities



Future Enhancements



Sentiment Analysis Integration

Link social media sentiment patterns with election data for comprehensive public opinion tracking



Mobile Dashboards

Develop responsive, touch-optimized interfaces for seamless mobile access and engagement



Local Elections Expansion

Extend coverage to municipal elections, school boards, and local referenda for comprehensive civic data



Predictive Analytics

Implement machine learning models to forecast outcomes and identify emerging trends before election day

Thank You!

